



CSC205 ARTIFICIAL INTELLIGENCE

Lecture # : 01

Topic: Application of AI, Latest Trends of AI and Future of AI

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**Content Creation
(Sora 2)**

🌟 Back at it, Nick

Hi Claude. We're looking at acquiring Joakim Poles for \$50M. Can you help me figure out the answers to my boss's questions, ASAP?

Research

Score: 4.5

AI chatbots

Physical Robots





Apple Intelligence



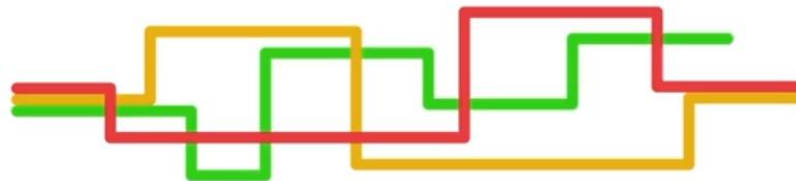
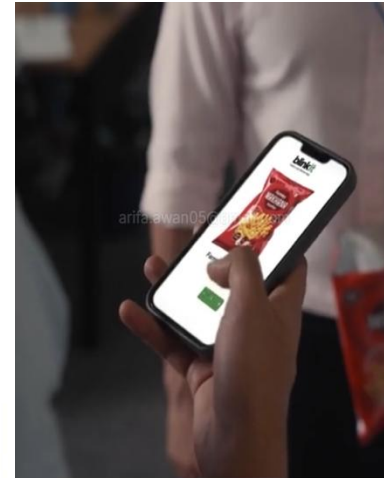
Movies Recommendation



Google Map



Travel time estimation



Travel time estimation



TECHNICAL TERM

AI is the branch of computer science that enable system to actually perform task that typically **required human intelligence.**



Speech Translation



Speech translation

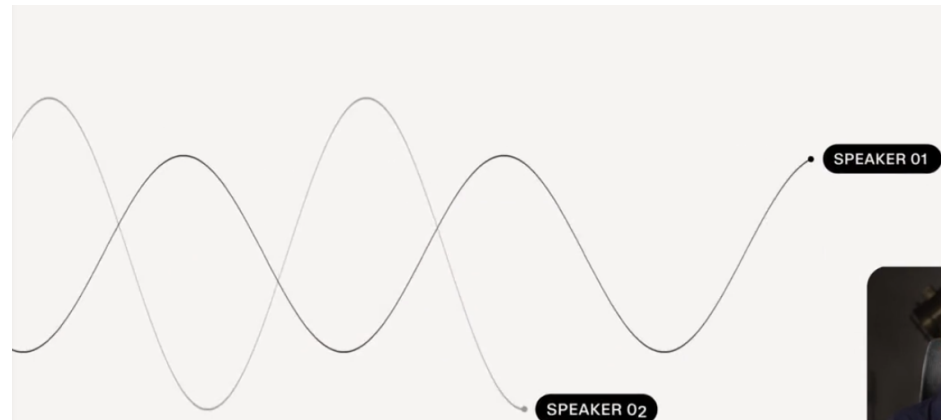
Scribe v2 Realtime sets a new benchmark for real-time transcription — outperforming every major streaming ASR service. Built on a new generation of models, it further advances the precision we introduced in Scribe v1.

Stream audio as it's captured and receive live transcripts

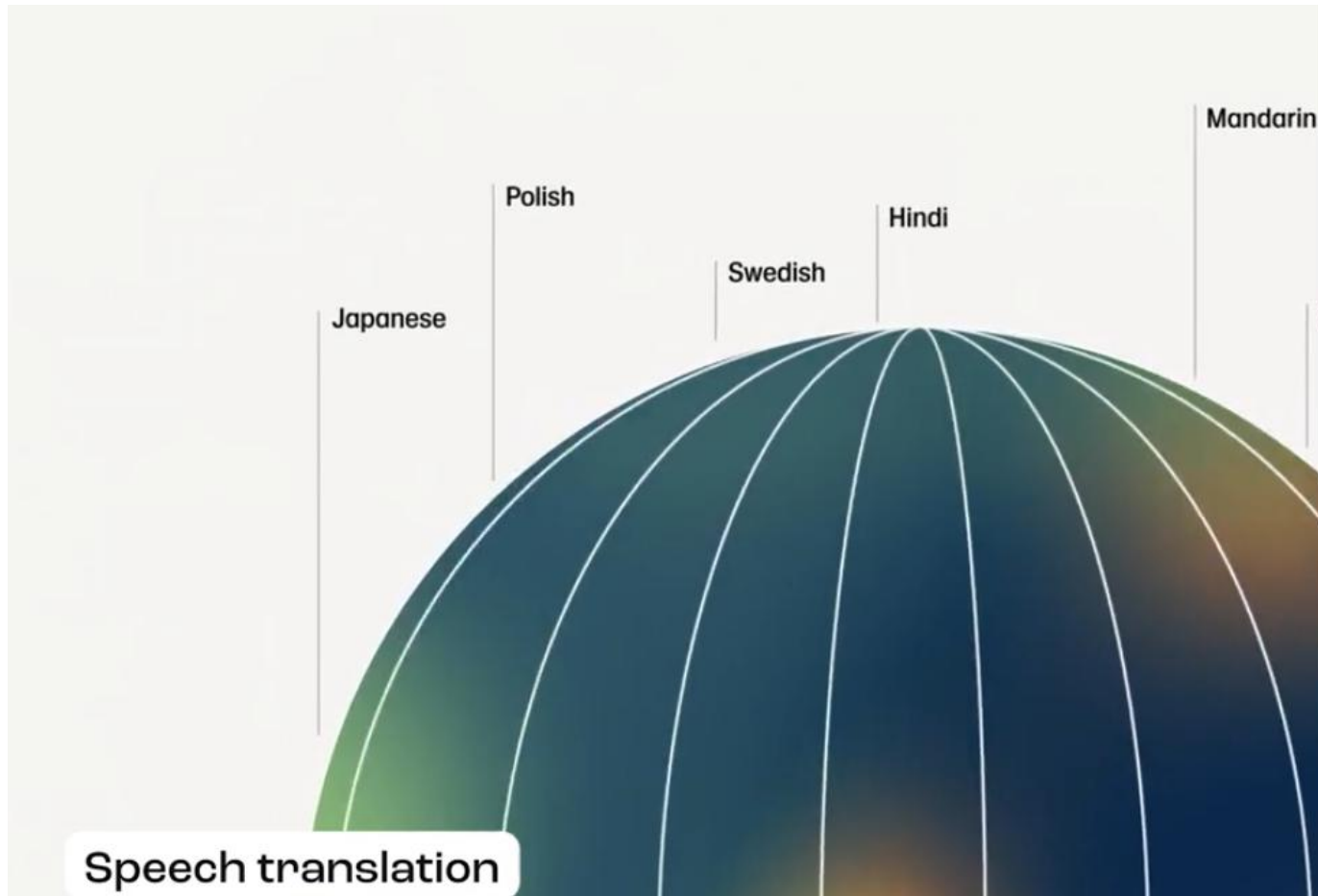
Speech translation



Speech translation



Speech Translation



Object Detection

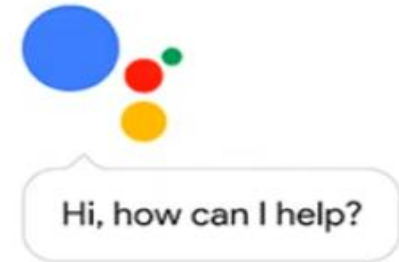
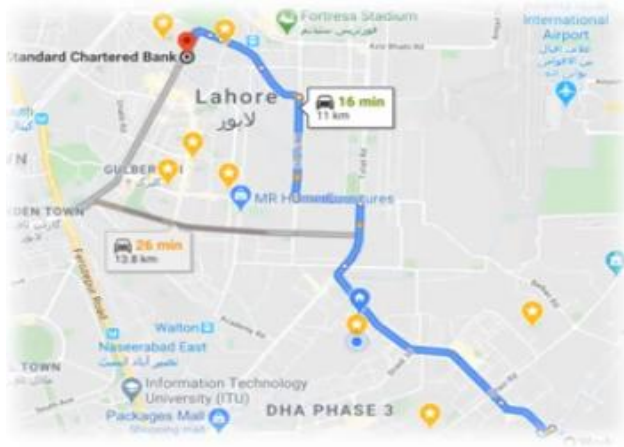




Google
UK

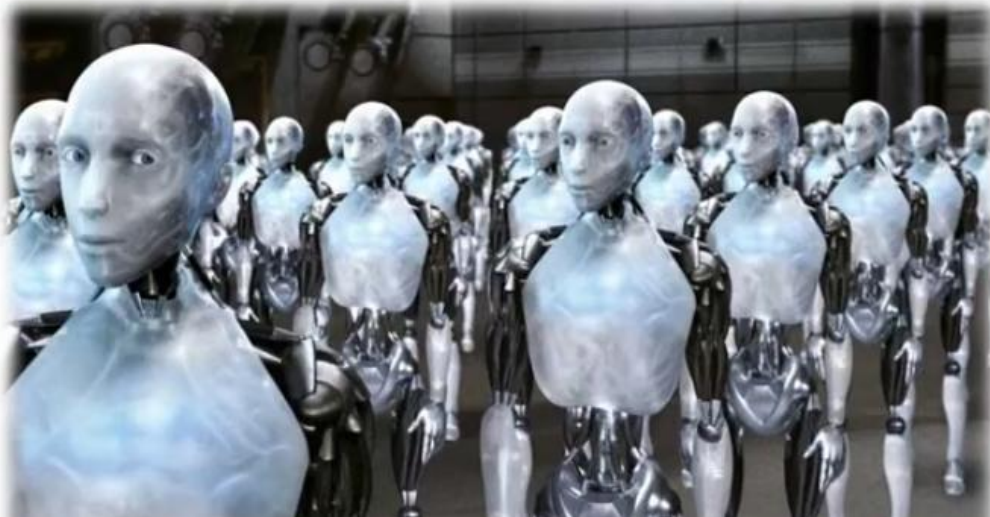


amazon alexa











Home / News / AI market projected to hit \$4.8 trillion by 2033, emerging as dominant frontier technology

AI market projected to hit \$4.8 trillion by 2033, emerging as dominant frontier technology

07 April 2025

AI's rapid rise risks deepening divides, with development concentrated in major economies and firms. Strategic investment and inclusive global governance are key to ensuring its benefits are widely shared.



AI takeover: Share of frontier tech market could quadruple by 2033

Estimated frontier technology market size and share of selected technologies, 2023 and 2033

Artificial intelligence Internet of things Blockchain Electric vehicles Low carbon steel Other



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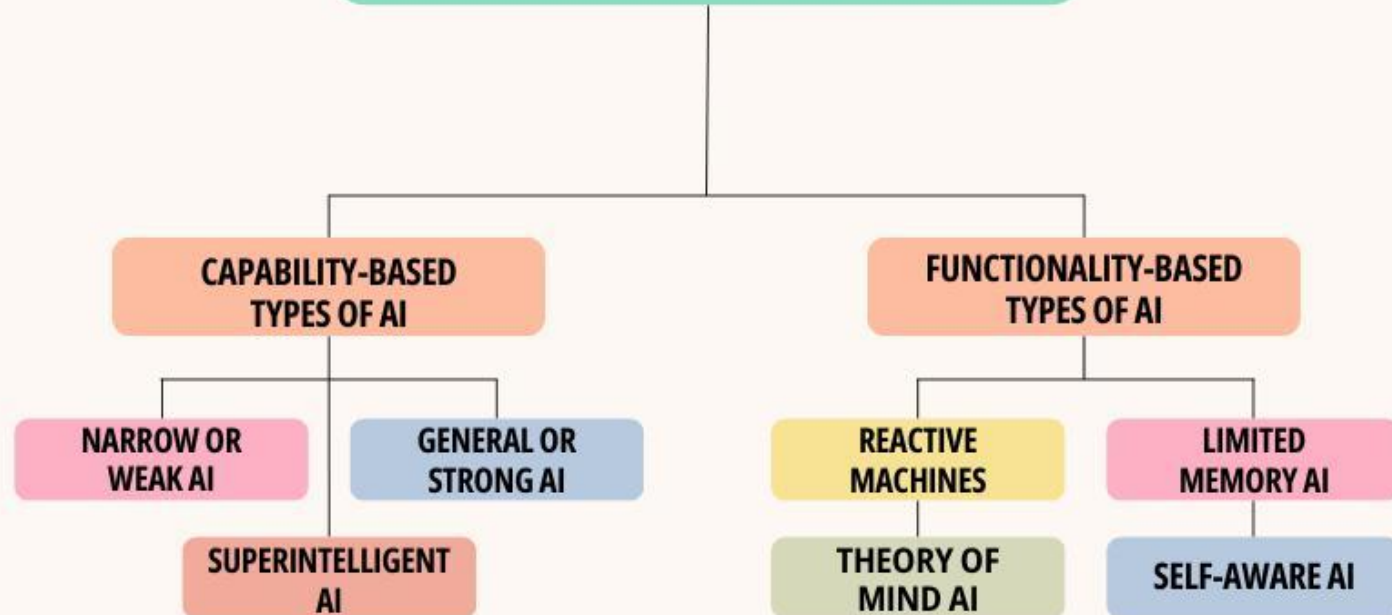
Stories

21 Oct 2025



From divides to dialogue, how's Asia's new economic order catching the AI boom?

TYPES OF ARTIFICIAL INTELLIGENCE



Capability Based AI (Types of AI)

ANI (Artificial
Narrow
Intelligence)

AGI(Artificial
General
Intelligence

ASI (Artificial
Super
Intelligence)



Artificial Narrow Intelligence

- AI designed to perform a specific task. It cannot think or learn beyond its programmed domain.
- Also called **Weak AI**.
- Specialized in one task or a limited set of tasks.
- Examples:
 - Chatbots
 - Voice assistants (Siri, Alexa, Google Assistant)
 - Recommendation systems (Netflix, YouTube)
 - Face recognition
 - Self-driving features
- **Example:** A chess AI can beat world champions at chess but cannot drive a car or write a novel.



Artificial General Intelligence

Also called: Strong AI

AGI refers to an AI system that could **learn, reason, adapt, and perform a broad range of intellectual tasks at a human level**, rather than being restricted to one specialized task. AGI remains a theoretical concept rather than an established existing technology



Artificial General Intelligence

Imagine an AI that could:

Learn mathematics → write software → teach a class → analyze a business → learn a new language → design a building without needing a separate AI system for every task.

- Status: ❌ No confirmed AGI currently exists.



Artificial Superintelligence.

ASI stands for **Artificial Superintelligence**.

It refers to a hypothetical AI whose intellectual capabilities would **surpass humans across essentially all relevant cognitive domains**. IBM classifies ASI as theoretical.

Status: ✖ Theoretical.



Artificial Superintelligence.

- **Potential capabilities could include:**
 - Extremely advanced scientific reasoning
 - Faster problem solving
 - Advanced invention and discovery
 - Superior strategic reasoning
 - Rapid learning



Functionality Based AI.

**Reactive
Machine**

**Limited
Memory AI**

**Theory of
Mind AI**

**Self Aware
AI**

Reactive Machines

- These are the simplest type of AI.
- They:
- Respond to current input
- Do not retain memories of previous events
- Perform specific tasks
- Cannot use past experiences to improve future decisions
- IBM Deep Blue, the chess-playing system that defeated Garry Kasparov.
- **Easy example:**
 *Chess AI → looks at the current board → chooses a move*



Limited Memory AI

These systems can use **past or recently observed information** when making decisions.

Many modern AI applications are commonly described as limited-memory systems, including examples such as:

- **Self-driving systems**
- **Generative AI**
- **Virtual assistants**
- **Recommendation systems**

IBM notes that generative AI systems such as ChatGPT fall under limited-memory functionality.

Self-driving car → observes previous/current traffic information → decides when to brake, accelerate, or turn.



Theory of Mind AI

This is a **future/theoretical form of AI** that would understand things such as:

Human emotions

Beliefs

Intentions

Desires

is **not the same as achieving a true**

Theory of Mind.

Status: Not achieved.



Self Aware AI

This is the most hypothetical category.
A self-aware AI would theoretically have:

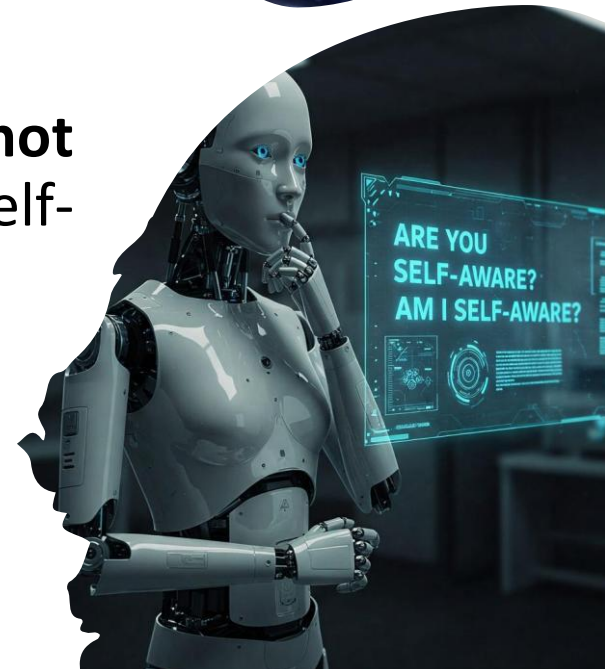
Awareness of itself

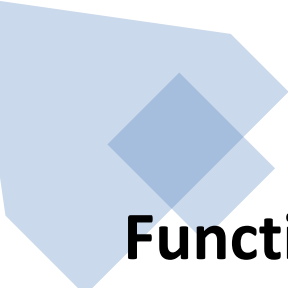
Understanding of its own internal state

Its own beliefs or desires

An understanding of itself as an entity.

Important: Current AI systems are **not** established to be conscious or self-aware. Self-aware AI remains theoretical.





Functional Type

Narrow AI?



Status

Reactive Machines

✓ Generally yes

Exists

Limited Memory

✓ Generally yes

Exists

Theory of Mind

✗ Not necessarily
/ theoretical

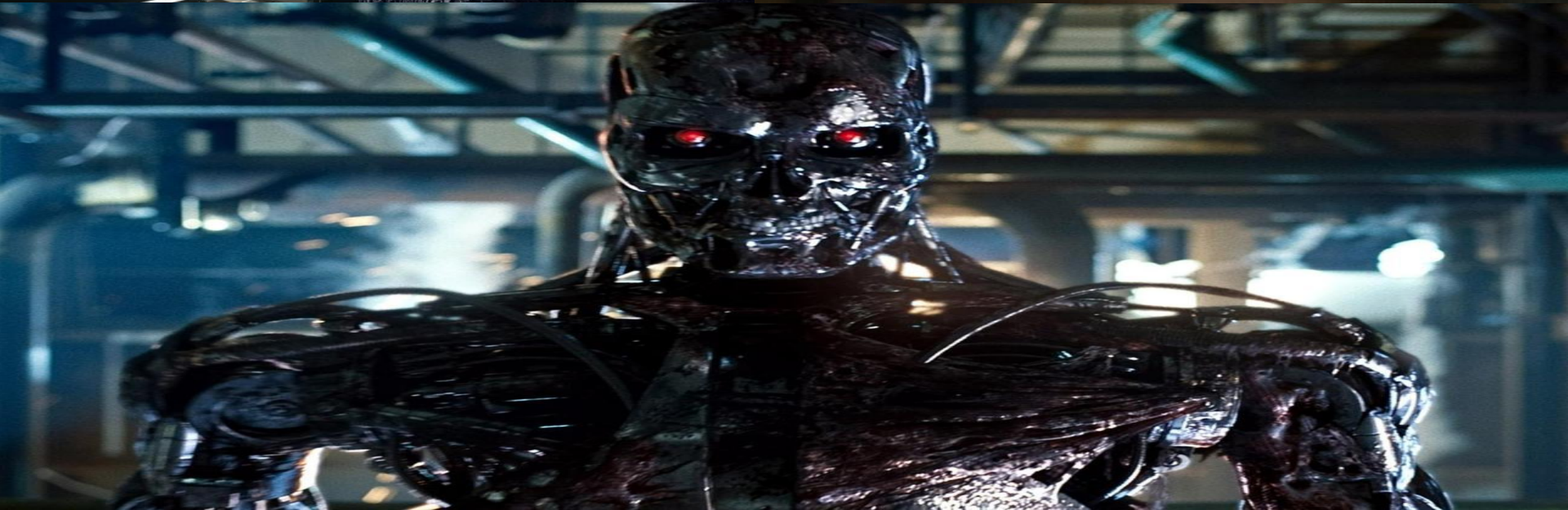
Not achieved

Self-Aware AI

✗ No — would
imply much more
advanced
intelligence

Theoretical





Movie	AI Character	Type it represents	Why?
Ex Machina	Ava	🧠 AGI / Self-Aware AI	General intelligence, emotions, manipulation, self-awareness
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Ex Machina	Ava	🧠 AGI / Self-Aware AI	General intelligence, emotions, manipulation, self-awareness
2001: A Space Odyssey	HAL 9000	🧠 AGI-like / Self-Aware AI	Reasoning, communication, planning, autonomous decisions
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Movies are not evidence that these AI types currently exist.

 Terminator's Skynet → fictional ASI

 Ex Machina's Ava → fictional AGI/Self-Aware AI

 Her's Samantha → fictional AGI/advanced Theory of Mind

These are **fictional representations** of future or hypothetical AI capabilities.



- **Today's AI:**

 ChatGPT, recommendation systems, face recognition → **mostly Narrow AI**

- **Science fiction:**

 Skynet, Samantha, Ava → **hypothetical AGI/ASI/self-aware AI**

Latest Trend of AI 2026

1. Agentic Ai

Agentic AI — AI That Can Take Action

Traditional AI gives you an answer.

Agentic AI can understand a goal, make a plan, use tools and complete the task.

Example:

You tell an AI:

“Plan my trip to Europe.”



Instead of only giving suggestions, an AI agent could:

Find suitable hotels

Check transportation

Compare prices

Create an itinerary

Prepare a schedule

Easy way to remember:

Chatbot → Gives answers

AI Agent → Gets things done



2. Multi-Agent AI

Instead of one AI doing everything, multiple AI agents work together, with each agent having a different responsibility.

Example: Online Shopping System



2. Multi-Agent AI

Imagine a team of AI agents:

Shopping Agent → Finds products

Price Agent → Compares prices

Review Agent → Checks reviews

Delivery Agent → Checks delivery

Manager Agent → Combines everything and makes the final recommendation

Easy way to remember:

One AI = One worker Multi-Agent AI = A team of AI workers



3. Digital AI Workforce — AI as a Digital

AI is increasingly being used like a digital worker to perform repetitive knowledge-based tasks.

Example:

A company receives hundreds of customer emails every day.



3. Digital AI Workforce — AI as a Digital

An AI digital worker can:

Read the email Understand the customer's problem

Check company records

Prepare a response

Forward difficult cases to a human It doesn't necessarily replace every employee;

instead, it can assist employees and automate routine work.



4. Physical AI

Physical AI combines AI + sensors + machines/robots so that AI can understand and act in the physical world.

A warehouse robot can:

- See objects using cameras
- Understand its surroundings
- Decide where to go
- Pick up packages
- Move them to another location



4. Physical AI

Autonomous vehicles

Humanoid robots

Smart manufacturing robots

Agricultural robots



4. Physical AI

- **Tesla China's Humanoid robot is installing AC Unit on Skyscraper.**
- **Robot do home chores.**
- **Robot making and serving the coffee.**
Outside Super Mall: Lujiazui, Shanghai
- **Dark Factory**

